

LCD and LCP of AG Codes

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Abstract. Linear complementary pairs (LCPs) of codes and linear complementary dual (LCD) codes have gained significant attention due to their applications in coding theory and cryptography. In this work, we construct explicit LCPs of codes and LCD codes from function fields of genus $g \geq 1$.

Keywords. AG codes, algebraic curves, function fields, LCD codes, LCP of codes.

Let $\mathbb{F}_q$ be the finite field of order q , where q is a power of a prime number. A linear code $\mathcal{C}$ over $\mathbb{F}_q$ is an $\mathbb{F}_q$ -subspace of dimension k in $\mathbb{F}_q^n$, where $n \geq 1$. A code $\mathcal{C}$ is said to have length n , dimension k , and (Hamming distance) minimum distance $d = d(\mathcal{C})$. These parameters are commonly denoted by $[n, k, d]$. For a given linear code $\mathcal{C}$, the dual code $\mathcal{C}^\perp$ is defined with respect to the Euclidean inner product in $\mathbb{F}_q^n$ and has dimension $n - k$. The hull of $\mathcal{C}$, with parameters $[n, k]$, is defined as

$$\text{Hull}(\mathcal{C}) = \mathcal{C} \cap \mathcal{C}^\perp.$$

The concept of the hull was first introduced by Assmus and Key in [1] for the classification of finite projective planes. If $\dim(\text{Hull}(\mathcal{C})) = \ell$, it follows that $0 \leq \ell \leq \min k, n - k$. In this case, $\mathcal{C}$ is referred to as an ℓ -dimensional hull code. Specifically, if $\ell = 0$, $\mathcal{C}$ is called an LCD code, and if $\ell = k$, the code $\mathcal{C}$ is said to be self-orthogonal, i.e., $\mathcal{C}^\perp \subseteq \mathcal{C}$.

A pair $(\mathcal{C}, \mathcal{D})$ of linear codes over $\mathbb{F}_q$ of length n is called an LCP of codes if $\mathcal{C} + \mathcal{D} = \mathbb{F}_q^n$ and $\mathcal{C} \cap \mathcal{D} = 0$. In the special case where $\mathcal{D} = \mathcal{C}^\perp$, the code $\mathcal{C}$ is an LCD code. LCD codes were first studied by Massey in 1964 [13], and in [14], he demonstrated the existence of asymptotically good LCD codes. In 2004, Sendrier [19] refined this result by proving that LCD codes achieving the asymptotic Gilbert-Varshamov bound exist.

Over the past decade, LCD codes and LCPs of codes have received increased attention due to their applications in cryptography. These include direct sum masking as a countermeasure against side-channel attacks (SCA) and fault injection attacks (FIA) [17, 5, 6], as well as enhancing the robustness of encoded circuits against hardware Trojan horses [18, 17].

In 2018, Carlet, Mesnager, Tang, Qi, and Pellikaan proved in [9] that for $q > 3$, any linear code over $\mathbb{F}_q$ is equivalent to an LCD code. This result implies that any set of parameters $[n, k, d]$ achievable by linear codes over $\mathbb{F}_q$ with $q > 3$ is also achievable by Euclidean LCD codes. Jin and Xing [11] showed that an algebraic geometry code over $\mathbb{F}_{2^m}$ ($m \geq 7$) is equivalent to an LCD code. More recently, the classification of LCD codes has been extended to include codes with higher hull dimensions (see [12]).

Numerous studies have focused on the characterization and construction of LCPs of codes and LCD codes (see, for instance, [14, 22, 7, 8, 20] and references therein). In the context of AG codes, Beelen and Jin [3] explicitly constructed several classes of MDS LCD codes over the rational function field. Mesnager, Tang, and Qi [15] investigated LCD AG codes and provided examples over the projective line, elliptic and hyperelliptic curves, and the Hermitian curve. A few years later, Bhowmick, Dalai, and Mesnager [4] established conditions for constructing LCPs of codes and presented illustrative examples.

Chen, Ling, and Liu [10] determined the dimension of the hull of Reed-Solomon codes using AG codes. In 2024, Moreno, López, and Matthews [16] introduced an explicit construction of non-special divisors of "small" degree over Kummer extensions. These divisors have been effectively used to describe LCD algebraic geometry codes from certain families over Kummer extensions.

In this work, we focus on the effective construction of LCPs of codes and LCD codes over function fields of genus $g \geq 1$ over $\mathbb{F}_q$. Our approach relies on the existence of non-special divisors of degree g and $g - 1$ supported on the rational places of the function field, as established in 2006 [2]. Theoretical conditions for non-special divisors to construct LCPs of codes over general function fields were provided in [4, Theorem 3.5]. These constructions depend heavily on the existence of non-special divisors of degree $g - 1$ in the function field. Moreover, for Kummer extensions, an explicit construction of such divisors was given in [16, Theorems 8, 12, 13], leading to the construction of several LCD codes over specific function fields.

References

- [1] E.F. Assmus Jr. and J.D. Key. Affine and projective planes. *Discrete Mathematics*, **83**(2-3):161–187, 1990.
- [2] S. Ballet and D. Le Brigand. On the existence of non-special divisors of degree g and $g - 1$ in algebraic function fields over $\mathbb{F}_q$. *Journal of Number Theory*, 116(2):293–310, 2006.
- [3] P. Beelen and L. Jin. Explicit MDS codes with complementary duals. *IEEE Transactions on Information Theory*, 64(11):7188–7193, 2018.
- [4] S. Bhowmick, D. K. Dalai, and S. Mesnager. On linear complementary pairs of algebraic geometry codes over finite fields. *Discrete Mathematics*, 347(12):114193, 2024.
- [5] J. Bringer, C. Carlet, H. Chabanne, S. Guilley, and H. Maghrebi. Orthogonal direct sum masking. In D. Naccache and D. Sauveron, editors, *Information Security Theory and Practice. Securing the Internet of Things*, 40–56, 2014.
- [6] C. Carlet and S. Guilley. Complementary dual codes for counter-measures to side-channel attacks. In R. Pinto, P. Rocha Malonek, and P. Vettori, editors, *Coding Theory and Applications*, 97–105, 2015.
- [7] C. Carlet, C. Guneri, F. Ozbudak, B. Özkaya, and P. Solé. On linear complementary pairs of codes. *IEEE Transactions on Information Theory*, 64(10):6583–6589, 2018.
- [8] C. Carlet, S. Mesnager, C. Tang, and Y. Qi. Euclidean and hermitian LCD MDS codes. *Designs, Codes and Cryptography*, 86(11):2605–2618, 2018.
- [9] C. Carlet, S. Mesnager, C. Tang, Y. Qi, and R. Pellikaan. Linear codes over $\mathbb{F}_q$ are equivalent to LCD codes for $q > 3$. *IEEE Transactions on Information Theory*, 64(4):3010–3017, 2018.
- [10] B. Chen, S. Ling, and H. Liu. Hulls of Reed-Solomon codes via algebraic geometry codes. *IEEE Transactions on Information Theory*, 69(2):1005–1014, 2023.
- [11] L. Jin and C. Xing. Algebraic geometry codes with complementary duals exceed the asymptotic Gilbert-Varshamov bound. *IEEE Transactions on Information Theory*, 64(9):6277–6282, 2018.

- [12] G. Luo, M-F. Ezerman, M. Grassl, and S. Ling. Constructing quantum error-correcting codes that require a variable amount of entanglement, *Quantum Information Processing*, **23** Article 4, 2024.
- [13] J. L. Massey. Reversible codes. *Information and Control*, 7(3):369–380, 1964.
- [14] J. L. Massey. Linear codes with complementary duals. *Discrete Mathematics*, 106-107:337–342, 1992.
- [15] S. Mesnager, C. Tang, and Y. Qi. Complementary dual algebraic geometry codes. *IEEE Transactions on Information Theory*, 64(4):2390–2397, 2018.
- [16] E. C. Moreno, H. H. López, and G. L. Matthews. Explicit non-special divisors of small degree, algebraic-geometric hulls, and LCD codes from Kummer extensions. *SIAM Journal on Applied Algebra and Geometry*, 8(2):394–413, 2024.
- [17] X. T. Ngo, S. Bhasin, J.-L. Danger, S. Guilley, and Z. Najm. Linear complementary dual code improvement to strengthen encoded circuit against hardware trojan horses. In *2015 IEEE International Symposium on Hardware Oriented Security and Trust (HOST)*, 82–87, 2015.
- [18] X. T. Ngo, S. Guilley, S. Bhasin, J.-L. Danger, and Z. Najm. Encoding the state of integrated circuits: a proactive and reactive protection against hardware trojans horses. In *Proceedings of the 9th Workshop on Embedded Systems Security*, 1–10, 2014.
- [19] N. Sendrier. Linear codes with complementary duals meet the Gilbert–Varshamov bound. *Discrete Mathematics*, 285(1):345–347, 2004.
- [20] L. Sok, M. Shi, and P. Solé. Constructions of optimal LCD codes over large finite fields. *Finite Fields and Their Applications*, 50:138–153, 2018.
- [21] H. Stichtenoth. *Algebraic Function Fields and Codes*. Graduate Texts in Mathematics. Springer Berlin Heidelberg, 2008.
- [22] X. Yang and J. L. Massey. The condition for a cyclic code to have a complementary dual. *Discrete Mathematics*, 126(1):391–393, 1994.